Signature Confidence Explorer

How many datasets stand behind a disease signature, and do they agree?

CMSE 830 Midterm Project | Shubham Koirala | Data: OCTAD 2.0

Most disease signatures rest on a single dataset

61%

of 350 diseases (213) are backed by one dataset only. They cannot be checked for replication or agreement.

Only 35% (123) have a gene backed by 6 or more.

USER

A researcher about to build on a disease signature

QUESTION

Is it backed by more than one study, and do the studies agree?

DATA

OCTAD 2.0 meta analysis: 350 diseases, 7.3 million gene results

ONE ROW

One gene in one disease

Three data decisions that changed the analysis

8.1%

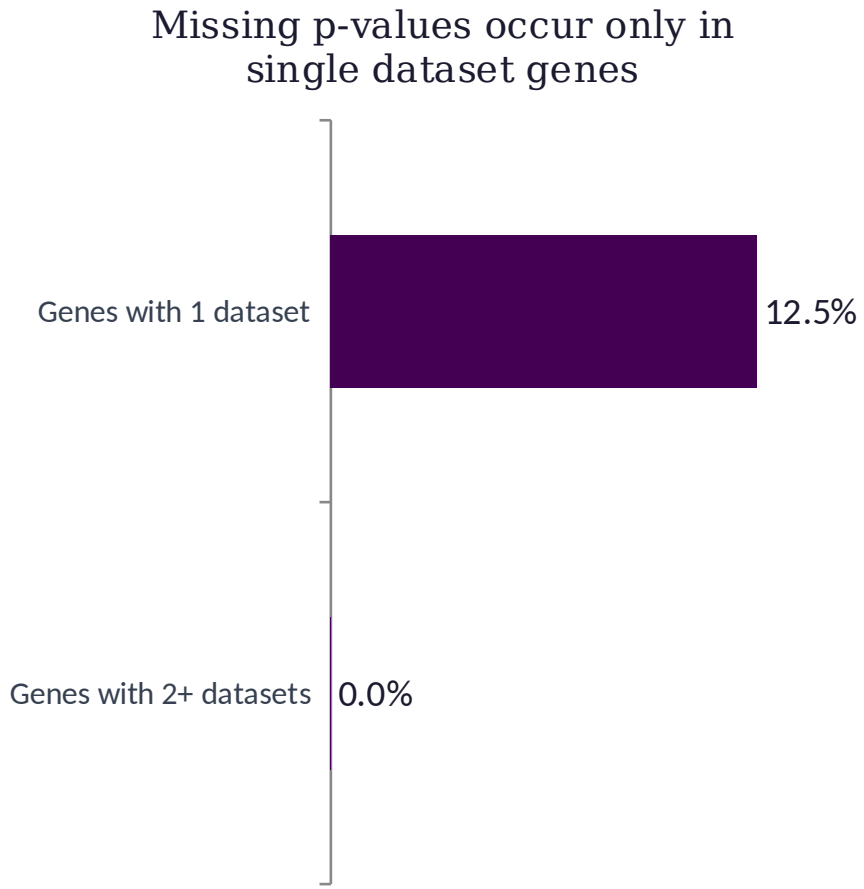
of adjusted p-values are missing
Not imputed. All are single dataset genes, so the gap is not random.

4.7M

I^2 values reported as 0
Set to missing. One study cannot show disagreement, and 0 reads as perfect agreement.

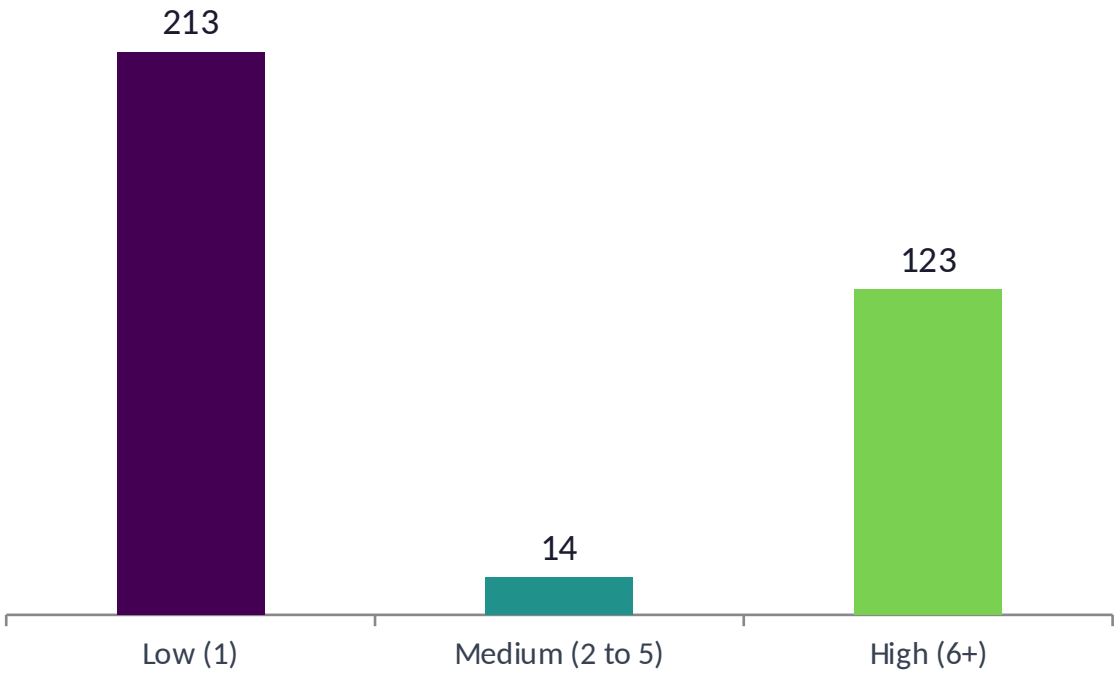
12,105

adjusted p-values exactly 0
Kept, plotted capped at 50. Genes on the cap are not ranked against each other.

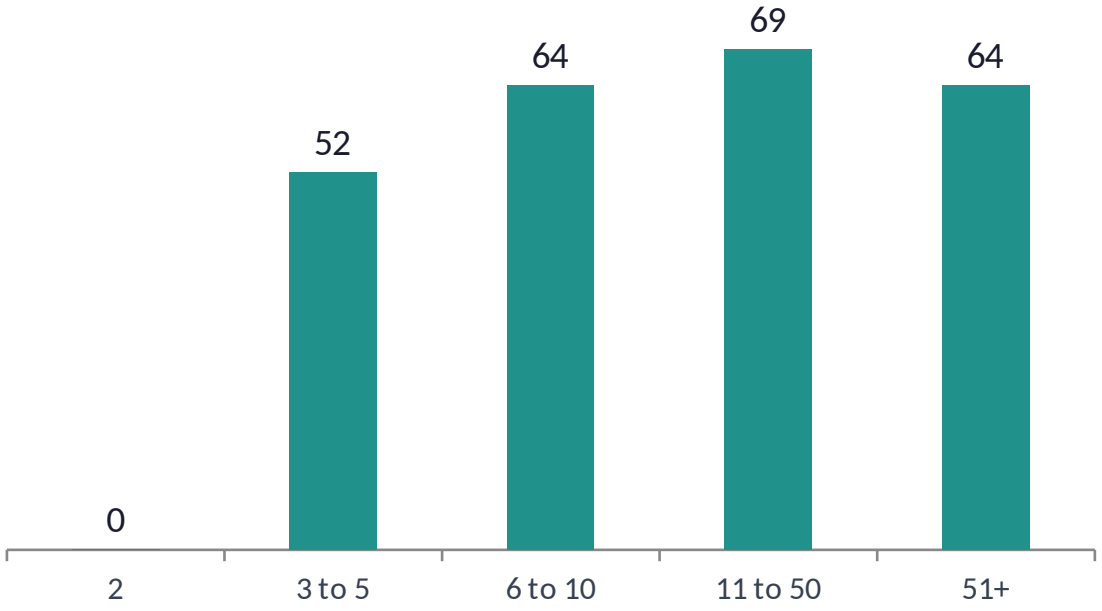


More datasets do not mean more agreement

Diseases by datasets behind their best gene



Median I^2 (disagreement, %) by datasets behind a gene



Across all genes, significance is unrelated to dataset count (rho 0.01) and effect sizes shrink (median |log2FC| 0.38 to 0.20).

The confidence card: same app, very different evidence

Gonorrhea

Confidence tier	Best supported gene	Significant genes (padj < 0.05)	Dataset disagreement (I^2)
Low	1 dataset	210	Not measurable
Low (1 dataset)	A typical gene has 1	1% of the genes that were tested	Needs 2 or more datasets

One dataset behind every gene. Nothing has replicated it, and agreement can't be measured.

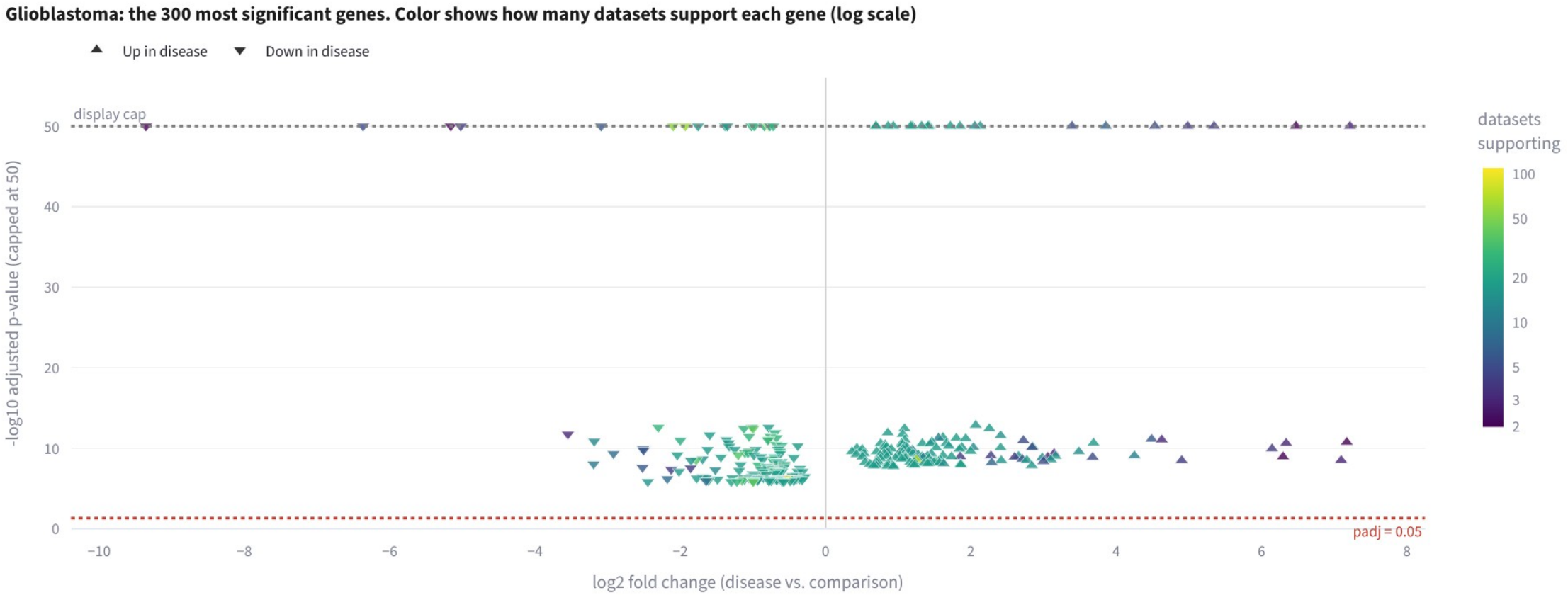
Glioblastoma

Confidence tier	Best supported gene	Significant genes (padj < 0.05)	Dataset disagreement (I^2)
High	110 datasets	7,386	Substantial
High (6 or more datasets)	A typical gene has 17	30% of the genes that were tested	Median 63%

Well supported: 66% of genes have 6+ datasets. Disagreement is substantial, so check each gene's I^2.

Pick a disease, read the tier and the I^2 label before looking at any gene.

The volcano plot: color shows dataset support



Slider hides genes with few datasets. Genes on the top dotted line are capped and not ranked against each other.

Limits and what comes next

LIMITS

- Dataset count is a proxy for confidence, not a measurement of it.
- k_datasets can reach the hundreds, so it may count comparisons, not studies.
- Only the 300 strongest genes per disease are plotted.
- Descriptive only. Tier cut points (1, 2 to 5, 6+) are my choice.

NEXT STEP

Add a pathway view of the replicated genes.

TRY IT

App link: add after deploying

AI use: Claude Sonnet 5.5 (via Claude Code) helped draft code, slides and text. Counts were re-checked against the full file; the decisions and interpretation are mine. Details in the README.